```
[1] import numpy as np
    import matplotlib.pyplot as plt
```

```
[2] # Fungsi f(x)
    def f(x):
        return (11 * x**3 - x + 80) / (x**2 + 1)

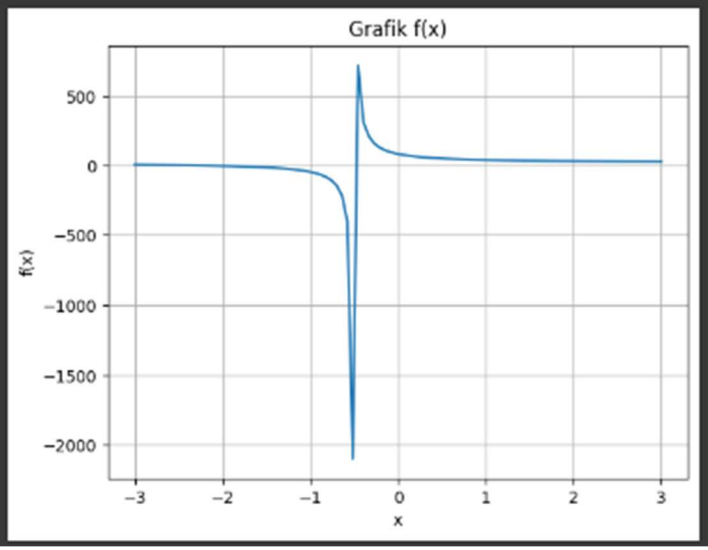
    # Soal nomor 1a: Hitung f(1) dan f(-1)
    x1 = 1
    x2 = -1
    f1 = f(x1)
    f_1 = f(x2)

    print("f(1) =", f1)
    print("f(-1) =", f_1)

    #Soal nomor 1b: Buat nilai x dari -3 sampai 3
    x = np.linspace(-3, 3, 100)
    y = f(x)
```

```
f(1) = 37.333333333333336
f(-1) = -48.0
```

```
plt.plot(x, y)
plt.title("Grafik f(x)")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)
plt.show()
```



[4]
```python
from sympy import symbols, limit, tan

# Definisikan variabel simbolik
x = symbols('x')
h = 2.6  # Nilai h berdasarkan NIM Anda

# Definisikan fungsi f(x)
f_x = (tan(x) - tan(h)) / (x - h)

# Hitung limit ketika x mendekati h
limit_f = limit(f_x, x, h)

print(limit_f)
```

```
oo
```

```python
print(f"Nilai h untuk NIM 4524210077 adalah: {h}")
print(f"Limit dari f(x) ketika x mendekati h adalah: {limit_f}")
```

```
Nilai h untuk NIM 4524210077 adalah: 2.6
Limit dari f(x) ketika x mendekati h adalah: oo
```

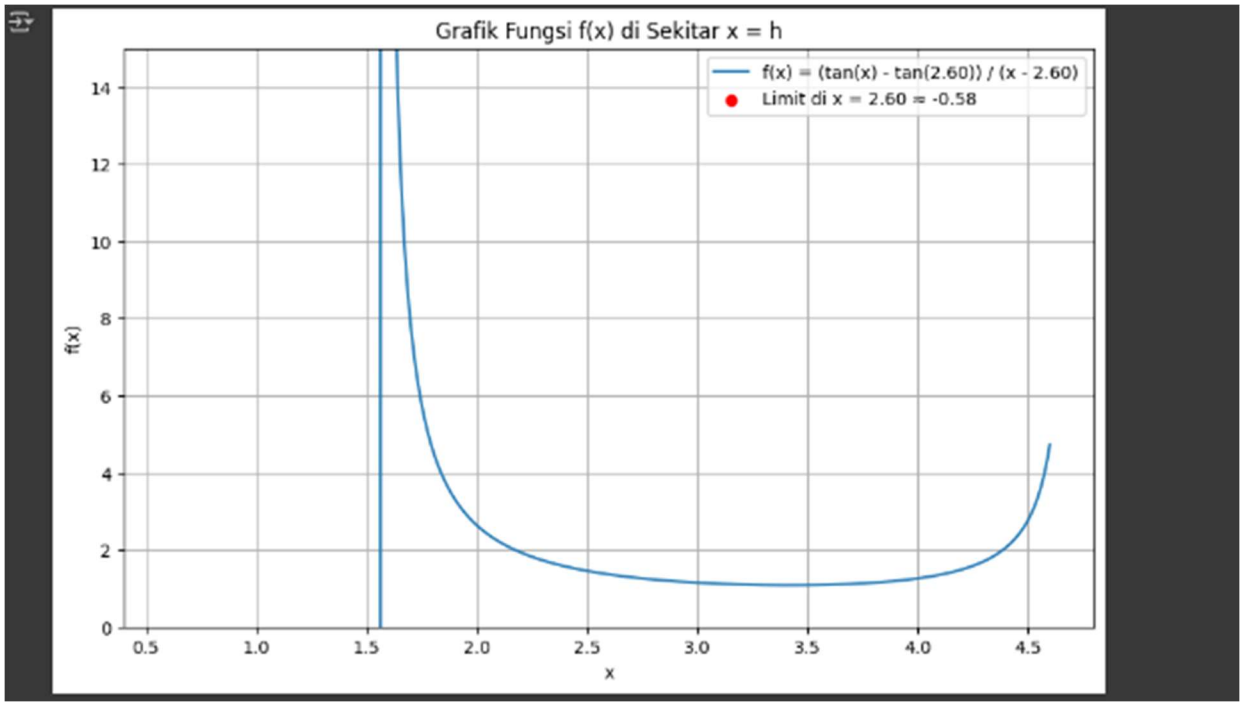
[7]
```python
import matplotlib.pyplot as plt
import numpy as np

h = 2.6

# Definisikan fungsi f(x)
def f(x):
  if np.isclose(x, h):
    return 1 / (np.cos(h)**2)
  else:
    return (np.tan(x) - np.tan(h)) / (x - h)

# Buat rentang nilai x di sekitar h
x_values = np.linspace(h - 2, h + 2, 400)
y_values = [f(x) for x in x_values]

plt.figure(figsize=(10, 6))
plt.plot(x_values, y_values, label=f'f(x) = (tan(x) - tan({h:.2f})) / (x - {h:.2f})')
plt.scatter(h, 1 / (np.cos(h)*2), color='red', label=f'Limit di x = {h:.2f} = {1 / (np.cos(h)*2):.2f}')
plt.title('Grafik Fungsi f(x) di Sekitar x = h')
plt.xlabel('x')
plt.ylabel('f(x)')
plt.grid(True)
plt.legend()
plt.ylim(0, 15)
plt.show()
```



https://colab.research.google.com/drive/1ko4oRdIaRGKrS4tBM1TRg7o-rlkie0hp?usp=sharing